

ENVS 193 Final Script

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Set Up

[Link to GitHub repository](#)

Read in packages

```
library(tidyverse)
library(here)
library(janitor)
library(readxl)
library(ggplot2)
library(ggeffects)
library(flextable)
library(MuMIn)
library(DHARMA)
library(rstatix)
```

Read in data

```
# bringing in screen_time.csv (personal data) with read_csv function
#   and storing it as object called my_data
my_data <- read_csv(here("data/screen_time.csv"))

# bringing in nest_data_final.csv with read_csv function and storing it
#   as object called nest_data
nests <- read_csv(here("data/nest_data_final.csv"))
```

Problem 1. Research writing

a. Transparent statistical methods

In part 1 of the results section, they used a logistic regression. One can deduce this because the predictor variable of distance to water edge is continuous, and the response variable of whether an American Avocet uses a habitat is binary. In part 2 of the results sections, they used a chi squared test. One can deduce this because the results are comparing the relationship between two categorical variables: bird species and habitat type.

b. Suggestions for rewriting

Our results suggest that distance to water edge predicts the probability of American Avocet microhabitat use. In addition, we found evidence to support that there is a relationship between birds species (American Avocet, Foster's Tern, Black-necked Stilt) and the habitats they are found in (managed pond, non-tidal marsh, salt pond, tidal marsh, or tidal mudflat).

c. Further analysis

Algae concentration is another variable that could influence the probability of American Avocet microhabitat use as algae is another important primary producer in aquatic ecosystems that influences food availability. It is a continuous predictor variable measured in mg dry weight/L water, and it can be collected in 1 liter containers with the dry weight measured in a lab.

Problem 2. Data analysis

a. Clean and wrangle

```
# cleaning nests data and storing it as nests_clean
nests_clean <- nests |>
  # selecting for just three columns
  select(height_cm, case_control, ant_species) |>
  # filtering for only three ant species
  filter(ant_species %in% c("RRB","AB","BBR")) |>
  # creating a new column to store full scientific names for ants
  #   based on abbreviations
  mutate(species_name = case_match(
    ant_species,
    "RRB" ~ "Crematogaster mimosae",
```

```

  "AB" ~ "Crematogaster sjostedti",
  "BBR" ~ "Crematogaster nigriceps"
)) |>
# ordering the species level
mutate(ant_species = as_factor(ant_species),
       ant_species = fct_relevel(ant_species,
                                "AB",
                                "RRB",
                                "BBR"))

# display structure of nests_clean data
str(nests_clean)

```

```

tibble [270 x 4] (S3: tbl_df/tbl/data.frame)
 $ height_cm      : num [1:270] 392 310 513 154 324 ...
 $ case_control   : num [1:270] 1 0 0 0 0 1 0 0 0 1 ...
 $ ant_species    : Factor w/ 3 levels "AB","RRB","BBR": 2 2 2 2 1 2 3 2 1 3 ...
 $ species_name   : chr [1:270] "Crematogaster mimosae" "Crematogaster mimosae" "Crematogaster mimosae" ...

```

```

# display 10 random rows from the nests_clean data
slice_sample(nests_clean,
             n = 10,
             by = NULL,
             weight_by = NULL,
             replace = FALSE)

```

```

# A tibble: 10 x 4
   height_cm case_control ant_species species_name
   <dbl>      <dbl> <fct>      <chr>
1     300          0 RRB      Crematogaster mimosae
2     170          0 BBR      Crematogaster nigriceps
3     132          0 BBR      Crematogaster nigriceps
4      74          0 RRB      Crematogaster mimosae
5     128          0 BBR      Crematogaster nigriceps
6     152          0 RRB      Crematogaster mimosae
7     247          1 RRB      Crematogaster mimosae
8     123          0 RRB      Crematogaster mimosae
9     171          0 RRB      Crematogaster mimosae
10    136          0 RRB      Crematogaster mimosae

```

b. Response variable

For the nest occurrence response variable, a 1 indicates that an Acacia tree contained a nest, and 0 indicates that an Acacia tree was available for a nest.

c. Purpose of study

Tree height, one of the predictor variables, is a continuous variable. As tree height increases, the probability of nest occurrence would increase because the nests are further off the ground and harder to get to for predators, which reduces nest predation, which can increase lifetime fitness in birds (Discussion, paragraph 1). In addition, taller trees likely have more branches and leaves which can create a denser canopy with or without the help of *C. nigriceps* that can provide further concealment of bird nests from predators (Discussion, paragraph 2).

Ant species, the other predictor variable, is a categorical variable with the four ant species representing four categories. The presence of *C. nigriceps* and *C. mimosae* would increase nest occurrence as they are aggressive defenders of host trees and thus can reduce the risk of nest predation compared to *C. sjostedi* and *T. penzigi* which are less aggressive host tree defenders (Discussion, paragraph 1). In addition, *C. nigriceps* in particular can increase nest occurrence because its pruning of the apical buds results in denser canopies which can provide further concealment of bird nests from predators (Discussion, paragraph 2).

d. Table of models

Model number	height_cm	species_name	Interaction?
1	X	X	No interaction (+)
2	X	X	Yes Interaction (*)

e. Run the models

```
# running a generalized linear model using glm
# and storing as nest_model_1 object
nest_model_1 <- glm(
  # case_control as a response to height and species
  # with no interaction
  case_control ~ height_cm + species_name,
  # using nests_clean data
  data = nests_clean,
  # setting glm to binomial glm specifically
  family = "binomial")

# running a generalized linear model using glm
```

```
# and storing as nest_model_2 object
nest_model_2 <- glm(
  # case_control as a response to height and species
  # with interaction (using a * instead of +)
  case_control ~ height_cm * species_name,
  # using nests_clean data
  data = nests_clean,
  # setting glm to binomial glm specifically
  family = "binomial")
```

f. Select the best model

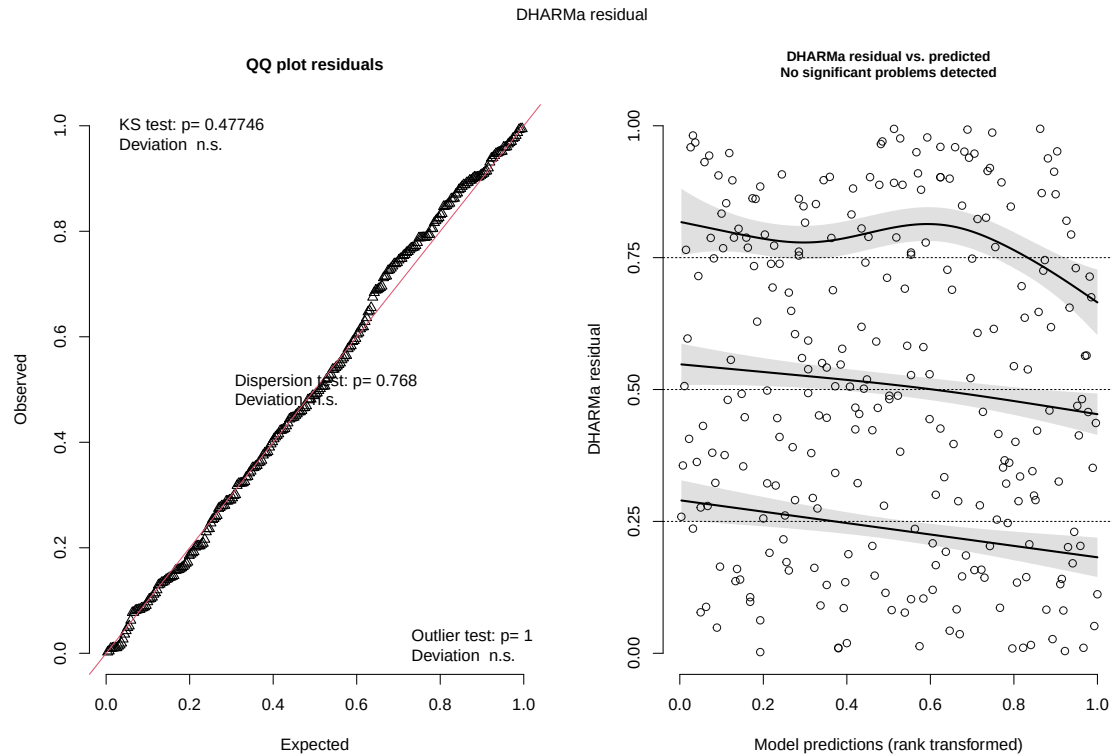
```
# AICc is Akaike's Information Criterion function
AICc(
  # running AIC for model 1 and model 2
  nest_model_1,
  nest_model_2
) |>
  # arranging the AIC scores by lowest at the top and highest at the bottom
  arrange(AICc)
```

	df	AICc
nest_model_2	6	238.1236
nest_model_1	4	258.8940

The best model as determined by Akaike's Information Criterion (AIC) had case factor as the response variable with tree height and ant species as interacting predictor variables.

g. Check the diagnostics

```
# checking diagnostics using simulated residuals function
plot(
  simulateResiduals(nest_model_2)
)
```



The points follow the QQ line relatively closely indicating that the residuals of the model follow a uniform distribution. The not significant result for the dispersion test tells us that our residuals are not more dispersed than expected from the uniform distribution mentioned in the prior sentence. The not significant result for the outlier test tells us that there do not appear to be any outliers that could affect the model coefficients or predictions. The plot on the right shows us that the homoscedasticity, or variance of residuals overtime, stay relatively constant over time. All of these results tell us that the data fits the model assumptions and thus we can use a logistic regression to predict values for the response variable of nest probability.

h. Visualize the model predictions

```
# creating model predictions and storing them as nest_predictions object
nest_predictions <- ggpredict(
  # model object
  nest_model_2,
  # predictors (in quotation marks)
  terms = c("height_cm[all]", "species_name")
) |>
# rename group column to species_name
```

```
# renaming will help with consistent terminology for the visualization
rename(species_name = group)
```

```
# ggplot is base layer
ggplot(data = nests_clean, # using nests_clean data
       mapping = aes(
         # x-axis is height
         x = height_cm,
         # y axis is case_control (nest probability)
         y = case_control,
         # coloring by species_name factor
         color = as.factor(species_name))) +
# changing ant colors manually
scale_color_manual(values = c("Crematogaster sjostedti" = "skyblue",
                              "Crematogaster mimosae" = "orchid3",
                              "Crematogaster nigriceps" = "springgreen3"))
) +
# changing ant fill colors manually
scale_fill_manual(values = c("Crematogaster sjostedti" = "skyblue",
                              "Crematogaster mimosae" = "orchid3",
                              "Crematogaster nigriceps" = "springgreen3"))
) +
# geom points is first layer
geom_point(
  mapping = aes(x = height_cm, # x-axis is acacia tree height
                y = case_control, # y-axis is nest occurrence
                color = species_name), # set to selected colors
  shape = 21, # change to open circle shape
  alpha = 0.6 # changing transparency
) +
# second layer is geom line, showing predictions
geom_line(data = nest_predictions, # data from model 2 prediction
          mapping = aes(x = x, # x-axis is height of acacia tree
                        y = predicted, # y axis is prediction
                          # of nest probability
                        color = species_name), # color by ant species
          linewidth = 1) +
# third layer is geom_ribbon to show 95% confidence interval
geom_ribbon(data = nest_predictions, # using data from model 2 prediction
           mapping = aes(x = x, # x-axis height of acacia
                         ymin = conf.low, # set low value
                         ymax = conf.high, # set high value
```

```

        fill = species_name), # fill color by ant species
    inherit.aes = FALSE, # ignore global aes
    alpha = 0.4) + # changing transparency
# create a panel for each ant species
facet_wrap(~ species_name) +
# relabeling x- and y-axis
labs(x = "Height of Acacia Tree (cm)",
     y = "Predicted Probability of Nest Presence",
     color = "species_name", # color by ant species
     title =
       "Predicted Nest Presence by Ant Species and Acacia Height (cm)"
) +
# changing theme from default
theme_minimal() +
theme(
  # removing panel background
  panel.background = element_rect(),
  # removing panel grid lines
  panel.grid = element_blank(),
  # removing legend
  legend.position = "none")

```

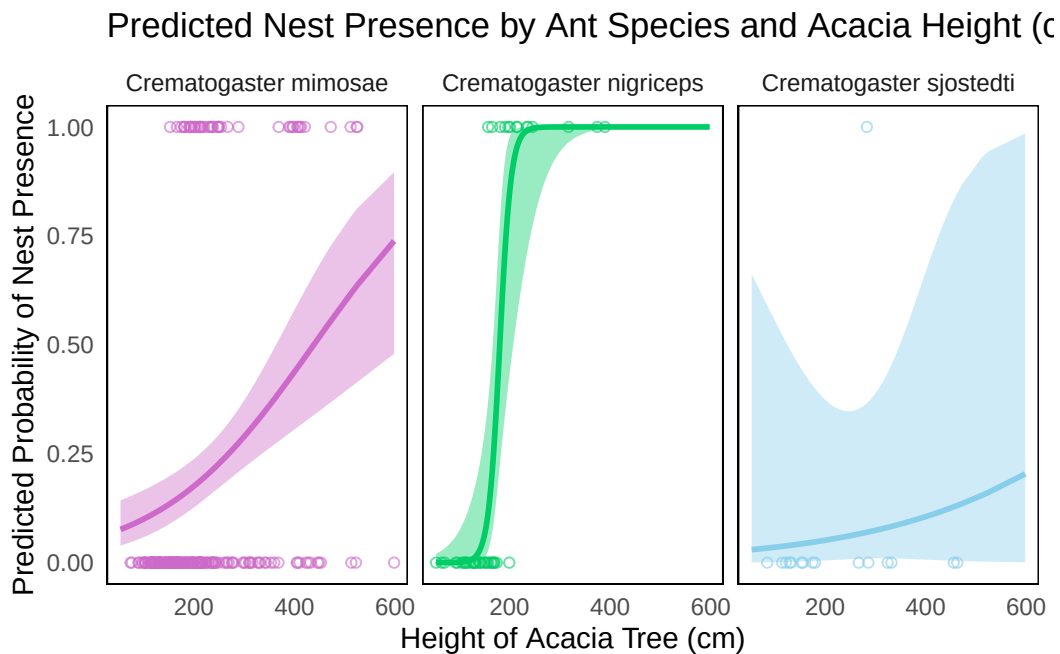


Figure 1: Predicted Probability of Bird Nest Occurrence Differs By Ant Species

and Tree Height There are three panels for the three ant species where purple represents *Crematogaster mimosae*, green represents *Crematogaster nigriceps*, and blue represents *Crematogaster sjostedti*. The x-axis is the acacia tree height, measured in centimeters. The y-axis is the predicted probability of the presence of a bird nest in an acacia tree. The open circle points represent actual observations of nest presence or absence in the acacia trees. The darker lines represent the model predictions of the probability of the nest presence for a given ant species and tree height. The ribbons above and below the dark lines represent a 95% confidence interval for the predicted probability of nest presence. Data originally from: Lujan, E., Nielsen, R., Short, Z., Wicks, S., Nderitu Watetu, W., Malingati Khasoha, L., Palmer, T. M., Goheen, J. R., & Alston, J. M. (2023). Data, code, and metadata for: Symbiotic acacia ants drive nesting behavior by birds in an African savanna [Data set]. Zenodo. <https://doi.org/10.5281/zenodo.8373322>

- i. Write a caption for your figure
Caption below the figure above.
- j. Calculate model predictions

```
# using ggpredict function to calculate
# predicted probability of nest occurrence for a given height
ggpredict(
  # using model 2
  nest_model_2,
  # calculating predicted nest probability with
  # a tree height of 600 cm as the first term
  # the species of ant is the second term
  terms = c("height_cm [600]",
            "species_name"))
```

```
# Predicted probabilities of case_control
```

```
species_name: Crematogaster mimosae
```

height_cm	Predicted	95% CI
600	0.74	0.48, 0.90

```
species_name: Crematogaster sjostedti
```

height_cm	Predicted	95% CI
600	0.20	0.00, 0.99

```
species_name: Crematogaster nigriceps
```

height_cm	Predicted	95% CI
600	1.00	1.00, 1.00

k. Interpret your results

At an Acacia Tree Height of 600 cm, the predicted probability of a nest occurrence is 0.2 (95% CI [0.00,0.99]) when *Crematogaster sjostedti* ants are present, 0.74 (95% CI[0.48,0.90]) when *Crematogaster mimosae* are present, and 1.00 (95% CI[1.00,1.00]) when *Crematogaster nigriceps* are present. Acacia tree height and ant species present interact to predict the probability of nest occurrence in Acacia trees. According to Figure 1, As Acacia tree height increases, the predicted probability of nest occurrence increases. Figure 1 also shows how both *Crematogaster mimosae* and *Crematogaster nigriceps* both had higher predicted nest occurrences than *Crematogaster sjostedti* likely because they are more aggressive defenders of the Acacia trees and by extension the bird nests. *Crematogaster nigriceps* in particular had the highest predicted nest occurrences because in addition to being an aggressive defender, *Crematogaster nigriceps* creates a denser canopy through the pruning of its trees which provides additional concealment of bird nests and thus increases the likelihood that the birds will nest in their trees.

Problem 3. Affective and exploratory visualizations

Cleaning my_data

```
# cleaning my_data and storing it as object my_data_clean
my_data_clean <- my_data |>
# dropping rows that contain NA values in the computer_time column
drop_na(computer_time) |>
# changing weekday variable inputs
mutate(weekday = case_when(
  # y inputs in the weekday column are now displayed as Weekdays
  # n inputs in the weekday column are not displayed as Weekends
  weekday == "y" ~ "Weekdays",
  weekday == "n" ~ "Weekends"
))
```

a. Comparing visualizations

One of the differences between my exploratory visualization and affective visualization is that the exploratory visualization contains more quantitative information on precise computer screen

time amounts between the weekdays and weekends. Nevertheless, my affective visualization is able to convey more emotionality on the differences in screen time through the water color background and chosen imagery for the weekday and weekend pseudo box plots.

A similarity between the two visualizations is that they both include box plots that can generally show the reader which group, weekdays or weekends, had more computer screen time. They also both had ways of conveying which box plot was for which group with the exploratory visualization using words, and the affective visualization using a weekly calendar image below the pseudo box plots to indicate weekdays vs weekends.

In the exploratory visualization I can see that weekdays have higher median computer screen times and more observations on the weekdays. For the affective visualization I can see that in general weekdays have greater screen time and that the maximum screen time recorded on the weekdays is greater than the maximum recorded screen time on the weekends.

In the week 9 workshop, my peers gave me the feedback to increase the emotional contrast between the weekdays and weekends even more. I implemented that suggestion by adding dark clouds above the weekday box plot, and adding a glimmer of light cast over the weekend box plot.

b. Designing an analysis

One analysis I could apply to answer my research question of how my computer screen time compares on weekdays vs weekends is a two-sample t-test. This would be an appropriate test to answer my question of how computer screen time compare on weekdays vs weekends as it would compare the average screen time between two sample groups: weekdays and weekends. It makes sense for the predictor variable as weekday vs weekend is a categorical predictor variables that can be used as the groups that I am comparing. It makes sense for the response variable of computer screen time as it is a continuous variable.

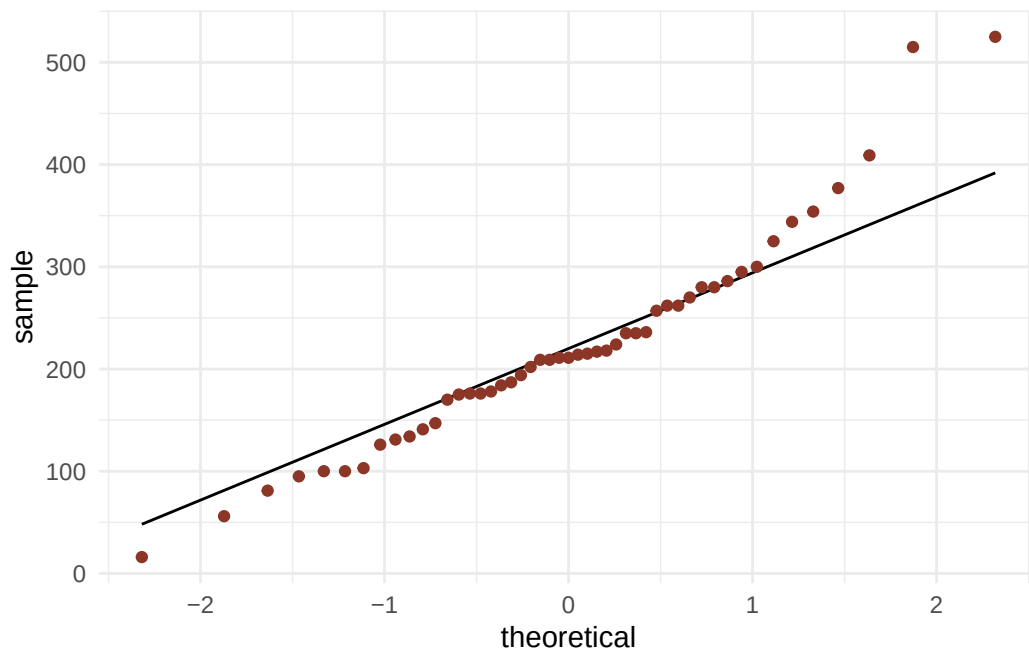
c. Check any assumptions and run your analysis

Checking that response variable is a continuous variable: Looking at my data I can confirm that computer screen time is a continuous variable measured in minutes.

Checking for independent observations: Measuring computer screen time on one day did not influence another day.

Check that the data is normally distributed:

```
# ggplot as base layer
ggplot(
  # using my_data_clean data
  data = my_data_clean,
  # plotting temperature
  mapping = aes(sample = computer_time)
) +
  # qq line is first layer
  geom_qq_line() +
  # qq points are second layer
  # changing color of the points
  geom_qq(color = 'tomato4') +
  # setting theme as minimal
  theme_minimal()
```



The qq points generally fall around the qq line with some tail off at the upper end. The computer_time response variable appears to be normally distributed overall. In addition, the central limit theorem also applies as there are more than 30 observations.

Checking for whether the variances are equal or not:

```
# using var.test to perform a F-test of equal variance
var.test(
```

```

# response ~ grouping
computer_time ~ weekday,
# using my_data_clean data
data = my_data_clean
)

```

F test to compare two variances

```

data:  computer_time by weekday
F = 1.5074, num df = 34, denom df = 13, p-value = 0.4341
alternative hypothesis: true ratio of variances is not equal to 1
95 percent confidence interval:
 0.5363613 3.4870201
sample estimates:
ratio of variances
      1.507375

```

The true ratio of variances is not equal to 1, and thus I will perform a Welch's two sample t-test.

Running my analysis: Welch's two sample t-test

```

# performing a t-test using t.test function
#   and storing it as my_data_t_test
my_data_t_test <- t.test(
  # response ~ grouping
  computer_time ~ weekday,
  # using cleaned data
  data = my_data_clean,
  # setting variances to be unequal
  #   since it is a Welch's two sampled t-test
  var.equal = FALSE
)

my_data_t_test

```

Welch Two Sample t-test

```

data:  computer_time by weekday
t = 1.4959, df = 29.326, p-value = 0.1454
alternative hypothesis: true difference in means between group Weekdays and group Weekends is
95 percent confidence interval:
 -16.24302 104.87159
sample estimates:
mean in group Weekdays mean in group Weekends
      234.0286           189.7143

```

Running a post-hoc test: Cohen's d

```

# running cohen's d and storing it as object called
# my_data_cohens_d
my_data_cohens_d <- cohens_d(
  # response ~ grouping
  computer_time ~ weekday,
  # using my_data_clean data
  data = my_data_clean,
  # not pooling standard deviations as the variances are not equal
  var.equal = FALSE
)

my_data_cohens_d

```

```

# A tibble: 1 x 7
  .y.      group1  group2  effsize    n1    n2 magnitude
* <chr>    <chr>   <chr>   <dbl> <int> <int> <ord>
1 computer_time Weekdays Weekends  0.452    35    14 small

```

d. Create a visualization

```

# calculating summary statistics to include in the visualization
# store my_data_clean summary data as the object my_data_summary
my_data_summary <- my_data_clean |>
  # group by the weekday vs weekend
  group_by(weekday) |>
  # begin summarizing both groups
  summarise(
    # calculate mean computer time
    mean_computer_time = mean(computer_time),
    # summarise number of observations for weekday vs weekend groups
    n = n(),

```

```

    # calculate standard deviation for computer time
    sd_computer_time = sd(computer_time)
  )

# display my_data_summary data frame
my_data_summary

# A tibble: 2 x 4
  weekday mean_computer_time     n sd_computer_time
  <chr>      <dbl> <int>      <dbl>
1 Weekdays    234.    35    107.
2 Weekends     190.    14     87.5

# base layer is ggplot
# data is my_data_clean
ggplot(data = my_data_clean,
  # weekday vs weekend group on the x axis
  # computer screen time on the y axis
  # coloring the points based on weekday vs weekend
  mapping = aes(x = weekday,
    y = computer_time,
    color = weekday
  )) +
# first layer is a jitter plot
geom_jitter(
  # width is set to 0.15 to tighten the horizontal jitter
  width = 0.17,
  # height is set to 0 so that there is no vertical jitter
  height = 0,
  # open circle point shape for jitter points
  shape = 21,
  # increasing transparency so that it is easier to see the mean
  alpha = 0.8
) +
# adding a point that is the mean computer screen time of each group
# using my_data_summary
geom_pointrange(data = my_data_summary,
  # points displayed based on weekday vs weekend grouping
  mapping = aes(x = weekday,
    # mean computer time is the y value
    y = mean_computer_time,

```

```

# upper bound is the mean plus
# a standard deviation
ymax = mean_computer_time + sd_computer_time,
# lower bound is the mean minus
# a standard deviation
ymin = mean_computer_time - sd_computer_time)) +
# manually setting colors for the treatment groups
scale_color_manual(values = c("Weekdays" = "darkblue",
                              "Weekends" = "skyblue")) +
# changing y axis label and title
labs(x = "",
      y = "Computer Screen Time (minutes)",
      title = "Comparing Computer Screen Time on Weekdays vs Weekends"
    ) +
# changing theme from default theme to theme_classical
theme_classic() +
# removing the legend
theme(legend.position = "none",
      # removing panel background
      panel.background = element_rect(),
      # removing panel grid lines
      panel.grid = element_blank())

```

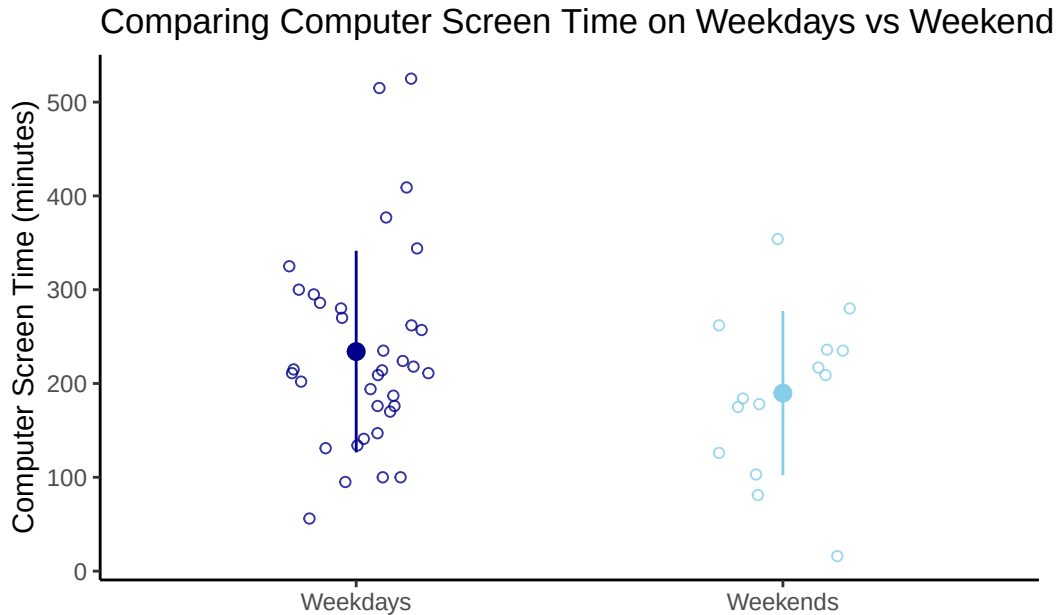


Figure 2: There is no statistically significant difference in computer screen time between Weekdays and Weekends Dark blue open points represent individual computer screen time observations on the weekdays whereas light blue open points represent individual computer screen time observations on the weekends. The filled in bigger points represents the mean computer screen time for both groups with the lines above and below the point representing one standard deviation above and below the mean. Data last updated: June 7, 2026.

e. Write a caption

Caption written above.

f. Write about your results

We did not find a significant difference in computer screen time between weekdays ($n = 35$, mean = 234 minutes) and weekends ($n = 14$, mean = 190 minutes; Welch's two sampled t-test, $t(29.3) = 1.50$, $p = 0.1454$, $\alpha = 0.05$).

We found a small (Cohen's $d = 0.452$) effect of the day of the week (Weekdays vs Weekends) on computer screen time.

g. Sharing your affective visualization

I attended the workshop for week 10! It was a nice touch to sign eachother's papers.